

Progressive Retinal Dystrophy Due To Retinol Transport Defect — Comprehensive Disease Report

Summary

Progressive retinal dystrophy due to a retinol transport defect is a rare inherited retinal degeneration caused primarily by **biallelic (autosomal recessive) loss-of-function variants in *RBP4***, the gene encoding **serum retinol-binding protein 4** (chromosome 10q23.33). RBP4 is the sole specific plasma carrier that mobilizes vitamin A (all-*trans* retinol) from hepatic stores and delivers it to peripheral tissues, most critically the eye. When circulating RBP4 is absent or non-functional, the retina is starved of the retinoid chromophore required to regenerate visual pigment, even though liver vitamin A stores remain intact. The result is a **systemic ("inherited") vitamin A deficiency** with a retina-dominant phenotype: childhood-onset night blindness (nyctalopia), reduced visual acuity, and progressive rod-cone/retinal pigment epithelium (RPE) degeneration, frequently accompanied by iris and chorioretinal colobomata, severe comedogenic acne, and, in some pedigrees, developmental anomalies such as patent ductus arteriosus.

The distinctive **diagnostic biochemical signature** separates this disorder from ordinary dietary vitamin A deficiency: serum RBP4 is undetectable, serum retinol is very low (~1/6 of normal), retinyl esters are normal, transthyretin is normal, and serum retinol **fails to rise with oral vitamin A dosing** because there is no carrier to mobilize the vitamin from the liver. Electrophysiology shows reduced scotopic ERG responses and elevated dark-adaptation thresholds. Mechanistically related disease arises from defects in the retinol **uptake receptors *STRA6* and *RBPR2***, which extract retinol from RBP4 at blood-tissue barriers (RPE, choroid plexus). These genes define a broader "liver-eye vitamin A axis."

A key refinement discovered across the investigation is a **genotype-inheritance-phenotype dichotomy**: recessive, mainly truncating *RBP4* alleles produce **retinal degeneration**, whereas dominant, almost exclusively missense alleles produce **ocular malformations** (microphthalmia-anophthalmia-coloboma, MAC), with a maternal-effect component—consistent with the dual role of the retinoic-acid/vitamin-A pathway in both eye

morphogenesis and the visual cycle. Management is empirical high-dose vitamin A supplementation (rescue demonstrated in *Stra6*-knockout mice), with AAV gene therapy proposed given the small size of *RBP4*.

Key Findings

Finding 1 — *RBP4* is the causal gene; biallelic loss-of-function drives disease via vitamin A transport failure

Multiple independent consanguineous pedigrees establish recessive *RBP4* variants as the cause of progressive retinal dystrophy. Reported pathogenic alleles include a homozygous splice-site variant **c.111+1G>A** (PMID: 23189188), a homozygous variant **c.67C>T** (PMID: 32323592), and biallelic **c.248+1G>A** (PMID: 27892788). Across these families, affected patients had **undetectable serum RBP4**, severe vitamin A deficiency, and low serum retinol, with **normal transthyretin**—collectively establishing a **null / loss-of-function mechanism** rather than a transport-complex assembly defect.

"Exome sequencing identified a novel homozygous splice site variant (c.111+1G>A) in the gene encoding retinol binding protein 4 (RBP4)." — PMID: 23189188

"Both patients had undetectable levels of RBP4 in the serum suggesting that this mutation led to either mRNA or protein instability resulting in a null phenotype." — PMID: 23189188

"Bi-allelic mutations in RBP4 were identified (c.248+1G>A), consistent with a diagnosis of inherited vitamin A deficiency." — PMID: 27892788

Ontology anchors: Gene HGNC:9922 (*RBP4*); OMIM gene 180250. GO biological process: retinol transport (GO:0034633), retinoid metabolic process (GO:0001523).

Finding 2 — Characteristic multisystem phenotype

The clinical picture spans the eye and beyond. Ocular features include **night blindness and low vision from early childhood** with fundus findings typical of recessive retinitis pigmentosa (PMID: 32323592), and **retinal dystrophy combined with iris/chorioretinal colobomata** (PMID: 27892788, PMID: 23189188). Extraocular features include **severe**

childhood-onset acne vulgaris that segregates with the *RBP4* genotype (PMID: 32323592, PMID: 23189188) and **developmental abnormalities such as patent ductus arteriosus** (PMID: 23189188).

"presented with low vision and night blindness from early childhood" — PMID: 32323592

"confirming that mutations in *RBP4* segregated with the acne vulgaris phenotype in this family" — PMID: 32323592

"one patient exhibited developmental abnormalities including patent ductus arteriosus and chorioretinal and iris colobomas" — PMID: 23189188

Suggested HPO terms: Nyctalopia/night blindness (HP:0000662); Retinal dystrophy (HP:0000556); Rod-cone dystrophy (HP:0000510); Reduced visual acuity (HP:0007663); Iris coloboma (HP:0000612); Chorioretinal coloboma (HP:0000567); Acne (HP:0031287); Patent ductus arteriosus (HP:0001643).

Finding 3 — Mouse models recapitulate the dystrophy and demonstrate rescue by pharmacological vitamin A

Rbp4-deficient mice on a C57BL/6 background show **reduced ERG a- and b-wave amplitudes, loss of peripheral choroid and photoreceptor layer**, fewer ganglion cells and synapses, and developmental defects (retinal depigmentation, optic disc abnormality, persistent hyaloid artery). Their biochemistry mirrors humans: serum retinol was **undetectable while liver retinol accumulated** (PMID: 26974396). *Stra6*-knockout mice show markedly reduced ocular retinoids, choroid/RPE malformations, early cone death, and short rod outer segments—and, crucially, **high-dose vitamin A rescues vision** (PMID: 24852372).

"loss of the peripheral choroid and photoreceptor layer in the peripheral retinas" — PMID: 26974396

"accumulated retinol in the liver but it was undetectable in the serum" — PMID: 26974396

"treatment with pharmacological doses of vitamin A restored vitamin A transport across these barriers and rescued the vision of *Stra6*(-/-) mice" — PMID: 24852372

This is the single strongest piece of translational evidence supporting **empirical vitamin A supplementation** as therapy.

Finding 4 — Genetic heterogeneity of the retinol-transport axis (RBP4 → STRA6 / RBPR2)

The disorder is one node in a multigene "**liver–eye vitamin A axis**." RBP4 is the blood carrier; cellular uptake of retinol is receptor-mediated. **STRA6** is highly expressed in epithelia forming blood–tissue barriers (RPE, choroid plexus), and its loss causes reduced ocular retinoids, RPE malformation, and cone death ([PMID: 24852372](#)). A second systemic receptor, **RBPR2**, when knocked out in mice, shows decreased ocular retinoids and loss of visual function ([PMID: 35745101](#)). The axis is integrated in a recent review ([PMID: 41829974](#)).

"This receptor, identified as the Stimulated by retinoic acid gene 6 (Stra6) gene product, is highly expressed in epithelia that constitute blood-tissue barriers." — [PMID: 24852372](#)

"Blood transport of the lipophilic vitamin is mediated by the retinol-binding protein, RBP4." — [PMID: 24852372](#)

Related genes/ontology: *STRA6* (HGNC:30650), *RBPR2* (HGNC:34333), *TTR* (transthyretin, RBP4's stabilizing partner). GO cellular component: extracellular space (GO:0005615); apical plasma membrane (GO:0016324) for STRA6 at the RPE.

Finding 5 — The founding human family defines the diagnostic signature

Seeliger et al. 1999 ([PMID: 9888420](#)) described the first human *RBP4* patients: two affected sisters with **compound heterozygous missense mutations Ile41Asn (I41N) and Gly75Asp (G75D)**. Their labs defined the phenotype: all-*trans* retinol 0.18–0.19 μM (normal 0.7–1.5 μM) that **did not increase in a dose-response test**, RBP below detection threshold, and **normal retinyl esters**. Clinically: night vision problems, reduced acuity (20/25–20/40), discrete iris coloboma, "fundus xerophthalmicus" with RPE atrophy, elevated dark-adaptation thresholds, reduced scotopic ERG, abnormal EOG light rise, and acne.

"RBP was below detection threshold, and retinyl esters were normal." — [PMID: 9888420](#)

"did not increase in a dose-response test" — [PMID: 9888420](#)

"compound heterozygous missense mutations (Ile41Asn and Gly75Asp) in the gene for serum retinol binding protein (RBP)" — [PMID: 9888420](#)

The **failure of serum retinol to rise with oral vitamin A** is a pivotal treatment caveat: because RBP4 is required to mobilize hepatic retinol, systemic delivery may be limited, and the therapeutic effect (as in *Stra6* mice) likely depends on mass-action/pharmacological dosing rather than restoring the physiological carrier.

Finding 6 — Genotype–inheritance dichotomy with a maternal-effect component

Plaisancié et al. 2023 (PMID: 37586836), studying 7 new families / 13 patients, resolved the phenotypic spectrum into two modes:

"dominantly inherited, almost exclusively missense, associated with ocular malformations, in contrast to recessive, mainly truncating, associated with retinal degeneration" — PMID: 37586836

"The retinoic acid (RA) pathway plays a crucial role in both eye morphogenesis and the visual cycle." — PMID: 37586836

The dominant, malformation-causing alleles show **skewed (maternal) inheritance**—consistent with a maternal-effect mechanism whereby maternal RBP4 status influences retinoic-acid signaling during fetal eye morphogenesis.

Mechanistic Model / Interpretation

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variant in *RBP4*** (splice-site, nonsense, or destabilizing missense) → **leads to** absent or non-functional serum retinol-binding protein 4 (undetectable serum RBP4).
2. Absent RBP4 → **results in** failure to mobilize all-*trans* retinol from hepatic stores into the bloodstream; **liver retinol is retained/accumulates** while **serum retinol falls to ~1/6 normal** (demonstrated in humans and *Rbp4*-KO mice).
3. Low circulating retinol → **deprives** the retinal pigment epithelium of substrate for the **visual (retinoid) cycle**; STRA6/RBPR2-mediated uptake at the RPE has no ligand to extract.

4. Chromophore deprivation → **impairs** regeneration of 11-*cis*-retinal / rhodopsin in rod (then cone) photoreceptors → **manifests first as night blindness** (rods most retinoid-dependent).
5. Chronic chromophore starvation → **causes** progressive photoreceptor and RPE degeneration and peripheral choroid loss → **progressive retinal dystrophy / RP-like phenotype** with reduced ERG and elevated dark-adaptation thresholds.
6. **Branch (developmental / dominant-missense arm):** deficient retinoic-acid signaling during embryonic eye morphogenesis → **results in** ocular malformations (iris/chorioretinal coloboma; microphthalmia–anophthalmia–coloboma spectrum), influenced by maternal RBP4 status (maternal effect). *(Inferred from the RA-pathway role in morphogenesis rather than directly demonstrated in these pedigrees.)*
7. **Branch (systemic retinoid signaling):** low tissue retinoid tone in skin → **contributes to** dysregulated follicular keratinization and **severe acne vulgaris**; other developmental effects (e.g., patent ductus arteriosus) reflect retinoid-dependent organogenesis. *(Skin/PDA links are phenotypically associated and mechanistically inferred.)*

The liver–eye vitamin A axis (schematic)

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LIVER (retinyl ester stores)
  |  hydrolysis -> retinol
  v
[ RBP4 ] <-- stabilized by TTR --> BLOODSTREAM (retinol.RBP4.TTR)
  |                                     X <-- LESION: no RBP4 -> no carrier
  v
BLOOD-TISSUE BARRIER (RPE, choroid plexus)
  |  STRA6 / RBPR2 receptor uptake
  v
RPE -> visual (retinoid) cycle -> 11-cis-retinal
  |
  v
PHOTORECEPTORS (rods > cones): rhodopsin regeneration
  |  chromophore starvation
  v
Night blindness -> progressive rod-cone/RPE degeneration

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Upstream vs downstream

Level	Component	Role	Directionality
Initiating lesion	<i>RBP4</i> LoF variant	No plasma retinol carrier	Most upstream
Systemic	Serum retinol ↓, liver retinol retained	Substrate deprivation	Upstream
Barrier/ uptake	STRA6, RBPR2	No ligand to import	Intermediate
Tissue	RPE visual cycle	No chromophore regeneration	Downstream
End-organ	Rod → cone photoreceptors	Degeneration	Most downstream

Cell types and processes

- **Cell Ontology (CL):** retinal rod cell (CL:0000604); retinal cone cell (CL:0000573); retinal pigment epithelial cell (CL:0002586); retinal ganglion cell (CL:0000740); hepatic stellate cell (CL:0000632, retinoid storage).
- **UBERON anatomy:** retina (UBERON:0000966); retinal pigment epithelium (UBERON:0001782); choroid (UBERON:0002348); iris (UBERON:0001769); liver (UBERON:0002107).
- **GO biological processes:** visual perception (GO:0007601); retinoid metabolic process (GO:0001523); retinol transport (GO:0034633); retina development in camera-type eye (GO:0060041); photoreceptor cell maintenance (GO:0045494).
- **CHEBI chemical entities:** retinol / vitamin A (CHEBI:17336); all-*trans*-retinol (CHEBI:50211); 11-*cis*-retinal (CHEBI:16066); retinyl ester (CHEBI:63410); retinoic acid (CHEBI:26536).

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
9888420	<i>Phenotype in retinol deficiency due to hereditary RBP defect</i>	Human clinical	Founding family; diagnostic signature; missense compound-het; non-response to oral dosing (F5)
23189188	<i>Exome analysis: novel RBP4 mutation with retinal dystrophy + developmental abnormalities</i>	Human clinical/genetic	Splice variant c.111+1G>A; null phenotype; coloboma + PDA (F1, F2)
27892788	<i>Vitamin A deficiency due to bi-allelic RBP4 mutation</i>	Human clinical/genetic	Biallelic c.248+1G>A; inherited vitamin A deficiency (F1, F2)
32323592	<i>Homozygous c.67C>T RBP4 with RP and childhood acne</i>	Human clinical/genetic	Night blindness; acne segregation (F1, F2)
26974396	<i>Severe ocular phenotypes in Rbp4-deficient mice (C57BL/6)</i>	Model organism	Photoreceptor/choroid loss; serum retinol undetectable, liver retinol retained (F3)
24852372	<i>STRA6 critical for cellular vitamin A uptake</i>	Model organism/in vitro	STRA6 uptake receptor; vitamin A rescue in Stra6-KO (F3, F4)
35745101	<i>RBPR2-null mice: decreased ocular retinoids, visual loss</i>	Model organism	Second systemic receptor in the axis (F4)
41829974	<i>The Liver-Eye Axis of Dietary Vitamin A Homeostasis</i>	Review	Integrates axis mechanisms/receptors (F4)
37586836	<i>Clinical, genetic, biochemical signatures of RBP4 disease</i>	Human clinical/genetic	Genotype–inheritance dichotomy; maternal effect; RA pathway (F6)
34440435	<i>Leber congenital amaurosis genotype–phenotype</i>	Review (context)	Frames visual-cycle gene dystrophies and gene therapy landscape
17646742	<i>The eye and anorexia nervosa (dietary vitamin A deficiency)</i>	Human case (contrast)	

PMID	Title (abbrev.)	Evidence type	Supports
			Acquired vitamin A deficiency phenocopy; distinguishes dietary vs transport defect

Note on the anorexia case (PMID: 17646742): dietary hypovitaminosis A produces overlapping retinal dysfunction (impaired scotopic ERG, RP-like field constriction) but is corrected by supplementation and features prominent xerophthalmia—underscoring that the **inherited** transport defect is distinguished by undetectable RBP4, normal retinyl esters, and poor response to oral vitamin A.

Section-by-Section Knowledge Base Content

1. Disease Information

Inherited retinal degeneration from failed plasma vitamin A transport. **Synonyms:** RBP4-related retinal dystrophy; inherited/systemic vitamin A deficiency; retinol-binding protein deficiency; "fundus xerophthalmicus" (historical). **Identifiers:** Gene *RBP4* OMIM 180250; disease phenotype OMIM #615147 (RBP4-related retinal dystrophy / "Retinal dystrophy, iris coloboma, and comedogenic acne syndrome, RDCCAS"). Suggested MONDO: map to the RBP4-related inherited vitamin A deficiency / retinal-dystrophy branch. ICD-10 H35.5 (hereditary retinal dystrophy). Information is derived from **aggregated disease-level resources and small pedigree case series**, not large EHR cohorts.

2. Etiology

Primary cause: genetic—biallelic (recessive) LoF *RBP4* variants (retinal degeneration) or monoallelic dominant missense (ocular malformation). **Genetic risk factors:** consanguinity (most recessive families are consanguineous). **Modifier/environmental:** dietary vitamin A status may modulate severity; there are no established protective alleles. **Gene–environment interaction:** because the block is at transport, dietary vitamin A alone does not normalize serum retinol, but pharmacological loading may partially bypass the deficit (mass-action delivery).

3. Phenotypes

Night blindness (HP:0000662, early childhood, progressive, high frequency); retinal dystrophy/rod-cone dystrophy (HP:0000556/HP:0000510, progressive); reduced visual acuity (HP:0007663); iris coloboma (HP:0000612) and chorioretinal coloboma (HP:0000567, subset); acne (HP:0031287, severe, childhood-onset, variable); patent ductus arteriosus (HP:0001643, rare). QoL impact is dominated by progressive low vision and nyctalopia affecting mobility and independence.

4. Genetic/Molecular Information

Causal gene: *RBP4* (HGNC:9922; 10q23.33; OMIM 180250). **Variant classes:** splice-site (c.111+1G>A; c.248+1G>A), c.67C>T, and missense (I41N/G75D; dominant missense in MAC). **Classification:** pathogenic/likely pathogenic per ACMG for the recessive truncating alleles. **Functional consequence:** loss of function / null (undetectable serum RBP4) for recessive disease; the dominant missense alleles are proposed to act via altered retinoid signaling. **Modifier genes:** *STRA6*, *RBPR2*, *TTR* within the same axis. Epigenetic and chromosomal abnormalities: none established.

5. Environmental Information

Dietary vitamin A intake is the principal modifiable variable but cannot correct the transport block at physiological doses. No infectious agents. Acquired (dietary/malabsorptive) vitamin A deficiency is an important phenocopy for differential diagnosis.

6. Mechanism / Pathophysiology

See "Mechanistic Model" above for the full ordered causal chain, pathway map, and ontology anchors.

7. Anatomical Structures Affected

Primary organ: eye—retina (UBERON:0000966), RPE (UBERON:0001782), choroid (UBERON:0002348), iris (UBERON:0001769). **Secondary:** liver (retinoid storage; UBERON:0002107), skin (acne), cardiovascular (PDA). **Cells:** rods (CL:0000604), cones (CL:0000573), RPE (CL:0002586). **Subcellular:** RPE endosomal/visual-cycle machinery; photoreceptor outer segments. **Lateralization:** bilateral, generally symmetric.

8. Temporal Development

Onset: childhood (night blindness/low vision from early childhood). **Course:** chronic, slowly progressive rod–cone/RPE degeneration; developmental/malformation features are congenital. **Critical period:** embryonic eye morphogenesis (for coloboma/MAC via retinoic-acid signaling) and early postnatal photoreceptor maintenance.

9. Inheritance and Population

Inheritance: autosomal recessive (retinal degeneration; often consanguineous) vs autosomal dominant missense (ocular malformation, with maternal-effect skewing). **Prevalence:** ultra-rare; only a handful of families reported worldwide—no reliable population estimate. **Penetrance/expressivity:** variable expressivity across ocular and extraocular features. Founder effects not established; consanguinity is a strong contributor for recessive cases.

10. Diagnostics

Biochemistry (key): undetectable serum RBP4, very low serum retinol, **normal retinyl esters**, normal transthyretin; serum retinol **does not rise with oral vitamin A**. **Electrophysiology:** reduced scotopic ERG, elevated dark-adaptation thresholds, abnormal EOG light rise. **Imaging/fundus:** RP-like RPE atrophy, peripheral pigmentary change, coloboma. **Genetics:** WES/WGS or inherited-retinal-dystrophy gene panels including *RBP4* (and *STRA6*, *RBPR2*); single-gene confirmation of biallelic *RBP4* variants. **Differential:** dietary/malabsorptive vitamin A deficiency (normalizes with supplementation; RBP4 present), other recessive RP, Leber congenital amaurosis (visual-cycle genes; [PMID: 34440435](#)).

11. Outcome/Prognosis

Vision is progressively impaired; night blindness and low vision are the dominant disabilities. Not typically life-limiting (extraocular features—PDA—are exceptions). No large survival data; prognosis is driven by degree of retinal degeneration at diagnosis and potential responsiveness to vitamin A loading.

12. Treatment

Mainstay: empirical **high-dose (pharmacological) vitamin A supplementation** (NCIT: Vitamin A / Retinol Therapy). Rationale is strong preclinically—vitamin A rescued vision in *Strat6*-KO mice ([PMID: 24852372](#))—but the human treatment caveat is that serum retinol may not normalize because RBP4 is absent ([PMID: 9888420](#)); benefit likely depends on mass-action

delivery to the eye. **Supportive:** low-vision rehabilitation; dermatologic management of acne. **Investigational:** AAV-mediated *RBP4* gene replacement is a rational strategy given the small gene and the null mechanism (no approved therapy yet; no NCT identified in this investigation).

13. Prevention

Genetic counseling for consanguineous/at-risk families; **carrier and cascade testing;** prenatal/preimplantation options for known family variants. Ensuring adequate maternal vitamin A status is prudent given the retinoic-acid morphogenesis link, though it does not overcome the transport block.

14. Other Species / Natural Disease

Disease modeled and mechanistically dissected in **mouse (*Mus musculus*, NCBI:txid10090):** *Rbp4*^{-/-} (PMID: 26974396), *Stra6*^{-/-} (PMID: 24852372), *Rbpr2*^{-/-} (PMID: 35745101). Orthologs: mouse *Rbp4* (NCBI Gene 19662). No specific companion-animal natural disease was identified in this investigation.

15. Model Organisms

Mammalian genetic knockouts are the principal models. *Rbp4*^{-/-} mice recapitulate the human biochemical signature (serum retinol undetectable, liver retinol retained) and structural degeneration (photoreceptor/choroid loss, reduced ERG). *Stra6*^{-/-} mice add the therapeutic proof-of-concept (vitamin A rescue). **Limitations:** background-dependent severity (C57BL/6 shows severe ocular phenotypes); mouse retinoid handling and rod-dominance differ from human macula, so cone/macular outcomes may be imperfectly modeled.

Limitations and Knowledge Gaps

- **Ultra-rare disease, tiny N.** Conclusions rest on a small number of consanguineous pedigrees and case series; there are no cohort-level prevalence, penetrance, or natural-history data.
- **Treatment evidence is largely preclinical.** Vitamin A rescue is proven in *Stra6*-KO mice, but human treatment response—especially whether high-dose vitamin A halts or reverses

retinal degeneration—is not established in controlled studies, and serum retinol may not normalize.

- **Extraocular mechanism inferred.** The links from RBP4 loss to acne and to PDA are phenotypic associations with mechanistic inference (retinoid signaling) rather than direct experimental proof in these families.
- **Dominant-missense/MAC arm** is newer and less mechanistically resolved; the maternal-effect basis needs functional confirmation.
- **No approved gene therapy or registered clinical trial** was identified; AAV replacement remains conceptual.
- Some cited resources ([PMID: 41829974](#)) lacked accessible abstracts, limiting verbatim support.

Proposed Follow-up Experiments / Actions

1. **Prospective vitamin A dosing trial** in genetically confirmed *RBP4*-null patients with serial ERG, dark-adaptation, OCT, and serum retinol/RBP4 to quantify whether pharmacological loading slows degeneration and to define the effective dose.
 2. **AAV-*RBP4* gene-replacement proof-of-concept** in *Rbp4*^{-/-} mice (liver-directed vs intravitreal/RPE-directed) measuring serum retinol restoration and ERG rescue; compare with vitamin A loading.
 3. **Genotype–phenotype meta-analysis** aggregating all reported *RBP4* families to formalize the recessive-truncating (degeneration) vs dominant-missense (malformation) dichotomy and estimate expressivity/penetrance.
 4. **Mechanistic dissection of extraocular features**—retinoid profiling of skin and assessment of developmental cardiovascular phenotypes in *Rbp4*-KO models—to test the retinoid-signaling basis of acne and PDA.
 5. **Diagnostic algorithm validation:** prospectively confirm that the triad "undetectable RBP4 + low retinol + normal retinyl esters + non-response to oral vitamin A" reliably distinguishes inherited transport defect from acquired/dietary deficiency.
 6. **Registry/consortium** for *RBP4*, *STRA6*, and *RBPR2* patients to enable natural-history and future interventional studies.
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Evidence source legend: Human clinical/genetic — PMIDs 9888420, 23189188, 27892788, 32323592, 37586836. Model organism — PMIDs 26974396, 24852372, 35745101. Review/context — PMIDs 41829974, 34440435, 17646742.

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